Unit 3

Market Equilibrium

Objectives

After studying this unit, the student should be able to:

- Understand how the market attains equilibrium
- Understand the factors determine the market equilibrium

Market Equilibrium

Equilibrium refers to the market condition which, once achieved, tends to persist. In economics this occurs when the quantity of a commodity demanded in the market per unit of time equals the quantity of the commodity supplied to the market over the same time period. Geometrically, equilibrium occurs at the intersection of the commodity's market demand curve and market supply curve. The tendency to move toward the equilibrium price is known as the market mechanism, and the resulting balance between supply and demand is called a market equilibrium.

The price and quantity at which equilibrium exists are known, respectively, as the equilibrium price and the equilibrium quantity. Consumers bring demand to the market for buying goods to satisfy their wants. Producers or sellers bring supply of their goods to the market to sell them and earn profit. The market demand and supply determine prices of goods and services exchanged between buyers and sellers. Thus, market equilibrium is reached when market demand for and market supply of a good are equal and as a result, equilibrium prices and equilibrium quantities are determined. At such equilibrium, buyers find that they are able to buy exactly the same amount that they are demanding at the prevailing price and sellers are able to sell exactly the amount they are willing to supply at the prevailing price. In other words, there is no incentive for anyone in the market to change their behaviour. Thus, equilibrium is the condition, which once achieved tends persist in time.

When the market price is below the equilibrium price the quantity demanded exceeds the quantity supplied. At the price below equilibrium, buyers are willing and able to purchase an amount that is greater than the suppliers produce and offer for sale. The buyers will "bid up" the price by offering a higher price to get the quantity they want. The quantity demanded will fall while the quantity supplied rises in response to the higher price. On the other hand, when the market price is above the equilibrium price the quantity supplied exceeds the quantity demanded. That is when the price is too high the sellers have surplus with them. In order to sell the surplus units, sellers lower their price. As the price falls, the quantity supplied decreases and the quantity demanded increases (Neither demand nor supply are changed). So, at the equilibrium price the amount that buyers are willing and able to purchase is equal to the amount sellers produce and offer for sale.

Using a Graph to Determine the Equilibrium

By bringing together the market demand and supply schedules we can see how market forces determine equilibrium price and quantity of the good. The table 3.1 presents a hypothetical demand and supply schedules of commodity X.

Table 3.1 Demand and Supply Schedule

Price of X (in ₹)	Quantity Supplied (Q_S)	Quantity Demanded (Q_D)	Surplus (+) Shortage (-)	Pressure on Price
5	140	20	120	Downward
4	100	40	60	Downward
3	60	60	0	Equilibrium
2	40	80	-40	Upward
1	20	100	-80	Upward

When the price of commodity X is ₹1, buyers are willing and able to purchase 100 units but sellers are willing and able to offer only 20 units for sale. Therefore, there is a shortage of 80 units. At price of ₹5, buyers are willing and able to purchase only 20 units while sellers are willing to offer 14 units. Therefore, there will be a surplus of 120 units in the market. Let us now consider a price of ₹3. At this price, buyers are willing to purchase 60 units and sellers are willing to offer 60 units for sale. That is, at this price, there is neither a surplus nor a shortage.

Quantity supplied of commodity is equal to the quantity supplied. Thus P_X = ₹3 is the equilibrium price and $Q_S = Q_D$ = 6o is the equilibrium quantity.

At any other price other than the equilibrium price of ₹3, market forces are set in motion to raise or lower the price. At the prices above the equilibrium price, the quantity supplied exceeds the quantity demanded. For example, at P_X = ₹4, sellers are willing to put 100 units of commodity X on the market but buyers are willing to take only 40 units. There will be surplus or excess quantity supplied of the commodity. Then the sellers will attempt to dispose this surplus by lowering the price. As price falls, a greater quantity will be demanded. At lower prices sellers supply smaller quantities and buyers demand larger quantities until the equilibrium price of ₹3 is reached, at which the quantity supplied of 60 units of commodity X equals the quantity demanded and market clears.

On the other hand, at prices below the equilibrium price, the quantity supplied fall short of quantity demanded. For example, at P_X = ₹2, buyers are willing to purchase 80 units but sellers will be able to offer only 40 units. There is a shortage or excess quantity demanded. Unhappy with the shortage, and wanting more commodity X, buyers will bid up the price to induce sellers to supply them the desired amount. Then the sellers offer a greater quantity at higher prices. The price will again settle at P_X = ₹3, because at this price, the quantity demanded equals quantity supplied. Note that, price of ₹3 is the only price that will prevail in the market. There will be no tendency of this price to change. Such a price is referred to as equilibrium price and quantity traded or exchanged at this price is called equilibrium quantity. The market for the product is said to be in equilibrium when the quantity demanded equals the quantity supplied at a specific price.

The determination of equilibrium price and quantity can be shown graphically by bringing together the market demand and market supply curve on the same graph, as shown figure 3.1.

The intersection of market demand curve DD and market supply curve SS at point E defines the equilibrium price P* and the equilibrium quantity Q*. At the equilibrium price, quantity demanded is equal to the quantity supplied. Because there is no excess demand or excess supply there is no pressure for the price to change further.

Reaching equilibrium between demand and supply is the process of changes and adjustments which ultimately results in equilibrium price and quantity. Suppose that price is above the equilibrium level, say at P_1. At such higher price, there is excess supply or surplus of the commodity. Then the sellers would begin to lower prices in order to sell their excess suppliers.

This surplus is eliminated as prices fall, quantity demanded increases and quantity supplied would decrease until the equilibrium price P* is reached, at which quantity demanded = quantity supplied. The opposite will happen if the price is below the equilibrium price, say at P_O. There will be excess demand or shortage. Consumers are unable to purchase the entire commodity they want at below-equilibrium prices and they bid up the price. This would put upward pressure on price and quantity supplied increases and until price eventually reach the equilibrium price P*, and the market clears.

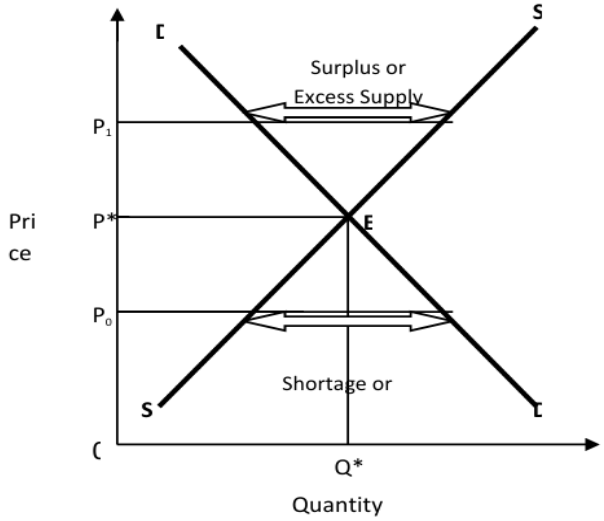


Figure 3.1: Market Equilibrium

Thus, through the process of adjustment in price and quantity, eventually equilibrium price and quantity are determined at which quantity demanded and supplied are equal. As long as demand and supply do not change, the equilibrium point remains the same. But in should be noted that, at a particular point in time, the observed market price may or may not be the equilibrium price. All we know is that market forces always push the market price towards the equilibrium price when they are not equal. We can also assume that, in the absence of price controls, the market price is the equilibrium price.

Using Math to Determine the Equilibrium

We can determine the equilibrium mathematically, using algebraic representations of the supply and demand curves. We use these two equations to solve for the equilibrium price at which the quantity demanded equals the quantity supplied (the equilibrium quantity).

The demand curve equation is given by

$$Qd = 160- 40p$$

The supply curve equation is given by

$$Qs = 50 + 15p$$

We want to find the equilibrium price, p, at which Qd = Qs = Q, the equilibrium quantity. Thus, we set the right sides of these two equations equal,

$$50 + 15p = 160- 40p,$$

and solve for the price. Adding 40p to both sides of this expression and subtracting 50 from both sides, we find that 55p = 110. Dividing both sides of this last expression by 55, we learn that the equilibrium price is p = 2. We can determine the equilibrium quantity by substituting this p into either the supply equation or the demand equation. Using the demand equation, we find that the equilibrium quantity is Q = 160- (40 × 2) = 80. We obtain the same quantity by using the supply curve equation: Q = 50 + (15 × 2) = 80.

Eg: There are 10,000 identical individuals in the market for commodity X, each with a demand function given by $Qd_x = 12- 2Px$, and 1000 identical producers of commodity X, each with a function given by $Qs_x = 20P_x$.

(a) Find the market demand function and the market supply function for commodity X.

(b) Find the market demand schedule and the market supply schedule of commodity X and from them find the equilibrium price and the equilibrium quantity.

(c) Plot, on one set of axes, the market demand curve and the market supply curve for commodity X and show the equilibrium point.

(d) Obtain the equilibrium price and the equilibrium quantity mathematically.

(a) $QD_x = 10,000 (12 - 2P_x)$

$= 120,000 - 20,000P_x$

$QS_x = 1000 (20P_x)$

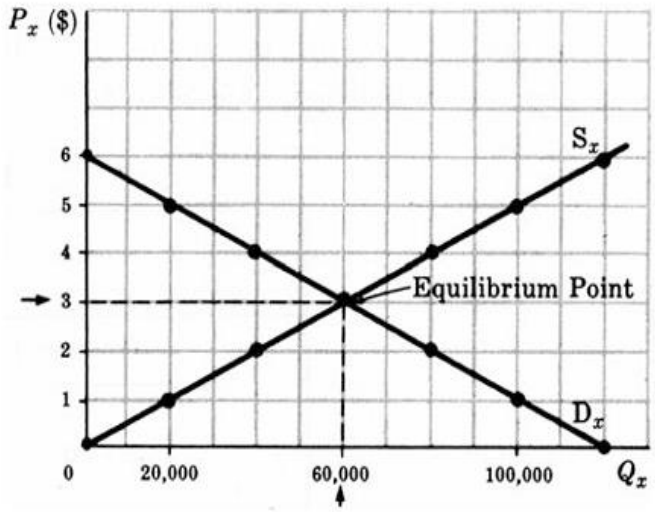
$= 20,000P_x$

(b) Equilibrium Table

Table 3.2: Equilibrium

Px ($)	QD_x	QS_x
6	0	120,000
5	20,000	100,000
4	40,000	80,000
3	60,000	60,000
2	80,000	40,000
1	100,000	20,000
0	120,000	0

(c) **Figure 3.2**



(c) $QD_x = QS_x$

$$120,000 - 20,000P_x = 20,000P_x$$

$$120,000 = 40,000P_x$$

P_x = \$3 (equilibrium price)

$QD_x = 120,000 - 20,000(3)$ or $QS_x = 20,000(3)$

= 60,000 (unit of X) = 60,000 (units of X)

Forces That Drive the Market to Equilibrium

If the market demand curve, the market supply curve, or both shifts, the equilibrium point will change. Ceteris paribus, an increase in demand (an upward shift) causes an increase in both the equilibrium price and the equilibrium quantity. On the other hand, given the market demand for a commodity, an increase in the market supply (a downward shift in supply) causes a reduction in the equilibrium price but an increase in the equilibrium quantity. The opposite occurs for a decrease in demand or supply. If both the market demand and the market supply increase, the equilibrium quantity rises but the equilibrium price may rise, fall or remain unchanged. The process of market adjustment can be visualized as changes in demand and/or supply.

a) Shifts of Changes in Demand

The demand function was defined from two perspectives:

1) A schedule of quantities that individuals were willing and able to buy at a schedule of prices during a given period, ceteris paribus.
2) The maximum prices that individuals are willing and able to pay for a schedule of quantities or a good during a given time period, ceteris paribus.

In both cases the demand function is perceived as a negative or inverse relationship between price and the quantity of a good that will be bought. The relationship between price and quantity is shaped by other factors or variables. Income, prices of substitutes, prices of compliments, preferences, number of buyers and expectations are among the many possible variables that influence the demand relationship. The demand function was expressed:

$$\mathbf{D_x = f\ (P_x, P_y, T, Y, A, N, Ep, U)}$$

Where, P_x = Price of the commodity, P_y = Price of related goods (including complementary and substitutes), T = Tastes and preferences of consumer, Y = Income level, A = Advertising and promotional activities, N = No. of consumers (Size of the market), Ep = Consumer's expectations about future prices and U = Specific factors affecting demand for a commodity such as seasonal changes, taxation policy, availability of credit facilities, etc. A change in the price of the good (P_x) is a change in quantity demanded or a movement along a demand

function. A change in any other related variable will result in a shift of the demand function or a change in demand.

In figure 3.3 the effects of a shift in demand are shown. If supply is constant, an increase in demand will result in an increase in both equilibrium price and quantity. A decrease in demand will cause both the equilibrium price and quantity to fall.

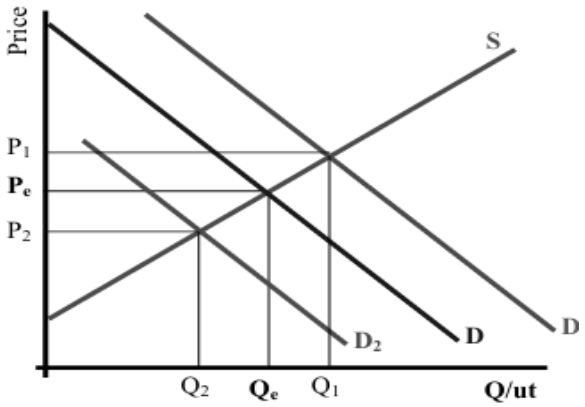


Figure 3.3 Shift in Demand

Given the supply (S) and the demand (D), the equilibrium price in the market is P_e. The equilibrium quantity is Q_e. An increase in demand is represented by a shift of demand from D to D_1. This will cause and increase in equilibrium price from P_e to P_1 and equilibrium quantity from Q_e to Q_1. A decrease in demand to D_2 will cause equilibrium price to fall to P_2 and quantity to Q_2.

b) Shift of Supply

The supply function was expressed,

$$\mathbf{S_X = f\ (P_X, P_Z, T, C, G_P)}$$

where, S_X = Supply of commodity X, P_X = Price of commodity X, P_Z = Price of related good, Z, T = Technological changes, C = Cost of production or price of inputs and G_P = Government policy or excise tax rate.

A change in the price of the good changes the quantity supplied. A change in any of the other variables will shift the supply function. An increase in supply can be visualized as a shift to the right, at each price a larger quantity is produced and offered for sale. A decrease in supply is a

shift to the left: at each possible price a smaller quantity is offered for sale. If the supply shifts and demand remain constant, the equilibrium price and quantity will be altered. An increase in supply (while demand is constant) will cause the equilibrium price to decrease and the equilibrium quantity to increase. A decrease in supply will result in an increase is the equilibrium price and a decrease in equilibrium quantity. It is shown in the figure 3.4.

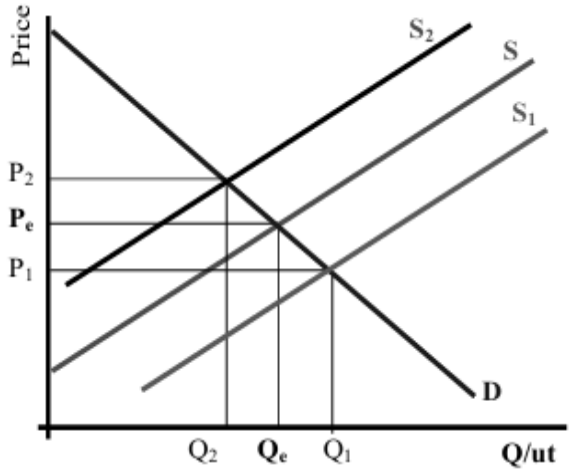


Figure 3.4 Shift in Supply

Given the demand (D) and the supply (S), the equilibrium price in the market is P_e. The equilibrium quantity is Q_e. An increase in supply is represented by a shift of supply from S to S_1. This will cause and decrease in equilibrium price from P_e to P_1 and an increase in equilibrium quantity from Q_e to Q_1. A decrease in supply to S_2 will cause equilibrium price to increase to P_2 and equilibrium quantity to fall to Q_2.

Changes in both Supply and Demand

When supply and demand both changes, the direction of the change of either equilibrium price or quantity can be known but the effect on the other is indeterminate. An increase in supply will push the market price down and quantity up while an increase in demand will push both market price and quantity up. The effect on quantity of an increase in both supply and demand will increase the equilibrium quantity while the effect on price is dependent on the magnitude of the shifts and relative structure (slopes) of supply and demand. The effect of an increase in both supply and demand is shown in figure 3.5.

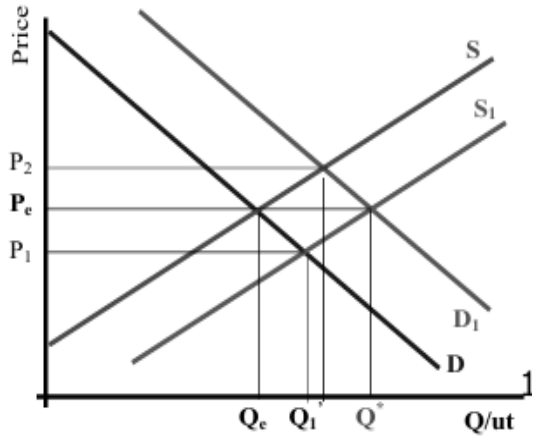


Figure 3.5: Shift in both Supply and Demand

Given supply (S) and demand (D), the equilibrium price is P_e and quantity is Q_e. An increase in supply to S_1 results in a drop-in price from P_e to P_1 while quantity increases from Q_e to Q_1. If demand increased to D_1, the equilibrium quantity would increase to Q*. The price however, is pushed up. In this case the price is returned to P_e. If the shift in demand were greater of less (or the slopes of S and D) were different, the equilibrium price might rise, fall or remain the same; the change is indeterminate until we have more information.

When supply and demand both shifts, the direction of change in either equilibrium price or quantity can be known but direction of change in the value of the other is indeterminate.

Multiple Choice Questions

1. A market will experience ---------- in a situation where quantity supplied exceeds quantity demanded and a ---------- in a situation where quantity demanded exceeds quantity supplied.

   (a) Shortage; shortage
   (b) Surplus; surplus
   (c) Shortage; surplus
   (d) Surplus; shortage

   Answer: (d)

2. The price of a good will tend to rise when
   (a) A temporary shortage at the current price occurs (assuming no price controls are imposed)

(b) A temporary surplus at the current price occurs (assuming no price controls are imposed)

(c) demand decreases (d) supply increases

Answer: (a)

3. Which of the following is true?

(a) The intersection of the supply and demand curves shows the equilibrium price and quantity in a market.

(b) A surplus is a situation where quantity supplied exceeds quantity demanded

(c) A shortage is a situation where quantity demanded exceeds quantity supplied.

(d) Shortages and surpluses ser in motion actions by many buyers and sellers that will move the market toward the equilibrium price and quantity unless otherwise prevented.

(e) All of the above are true.

Answer: (e)